MAPPING THE *Machine*

A Compensation, Tenure, and Structure Analysis of the District's Public Workforce

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MATH 014 — Introduction to Data Science
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The DC Public Employee Salary Dataset

What it is. Every full-time employee on the District of Columbia government payroll, published openly through DC Open Data.

Why it matters. Publicly funded payroll for a workforce larger than many Fortune 500 companies — the people who run the city we live in.

Variables
Name · Job Title · Agency · Grade · Annual Compensation · Hire Date · Appointment Type

40,611 raw rows

9 columns

88 agencies

3,698 job titles

What this project tries to answer

Three connected questions about how the DC government pays its people.

01 How does compensation vary across DC's agencies, and which agencies offer the highest median pay?

02 Does employee tenure predict pay — and does that relationship behave differently across appointment categories (CS, Ed, MSS, LS, Exe)?

03 Where is internal pay dispersion most extreme inside large agencies — and which agencies are most compressed?

Cleaning the Payroll

1 Strip Whitespace

2 Normalize Job Titles

3 Parse Hire Dates

4 $10K Floor (drop 4,984)

5 Fill 40 Missing GRADE → UNK

6 Engineer Features

7 Drop ID columns

After cleaning

35,627 rows · **12** columns · **0** missing critical fields

Engineered features: TENURE_YEARS · HIRE_YEAR · HIRE_DECADE · APPT_CATEGORY · APPT_STATUS

What we'll measure

Q1

Agency Comparison

Median pay by agency, with cross-tabulation against appointment category to expose mix effects.

Q2

Tenure–Pay Slope

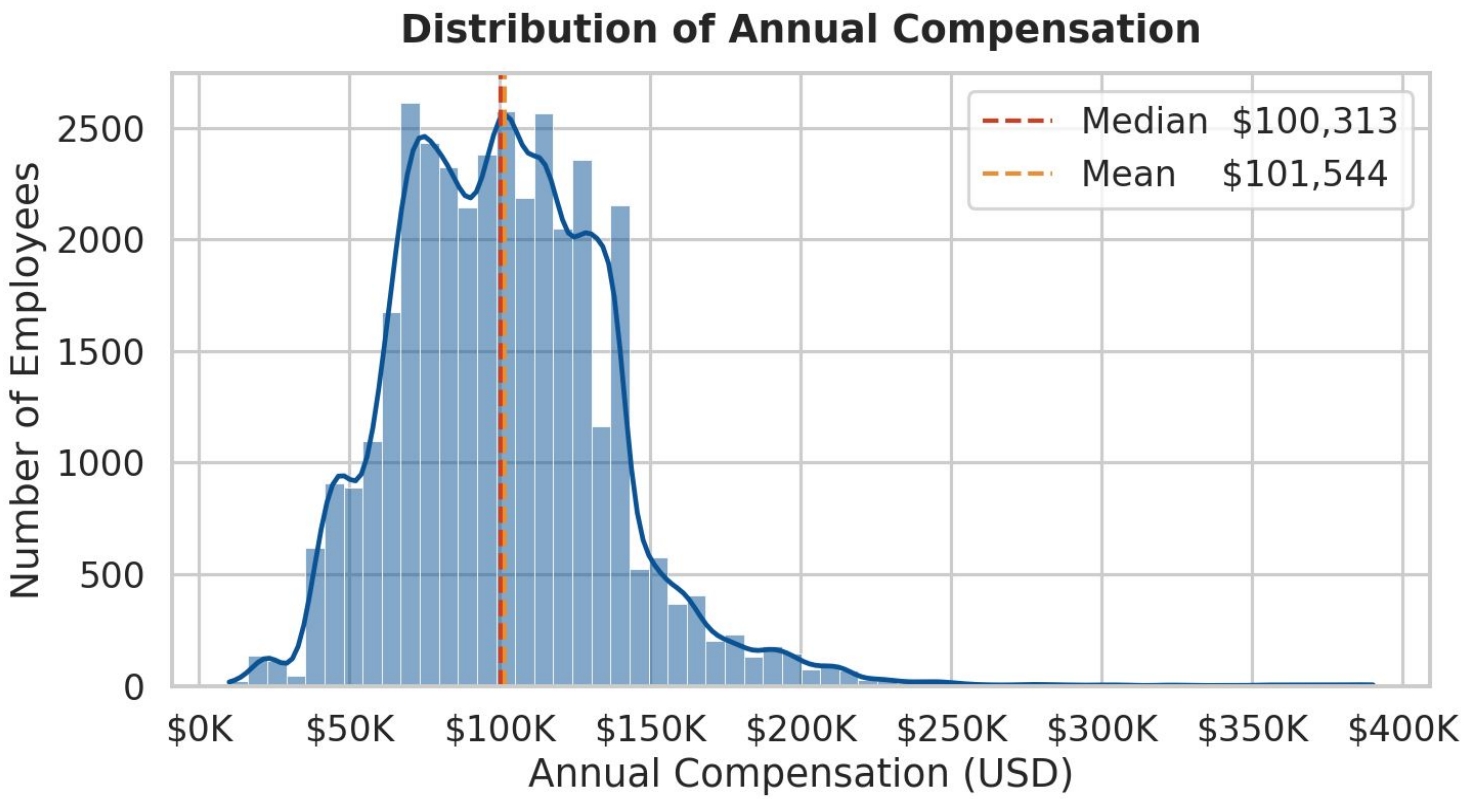
Pearson correlation overall, plus tenure-banded medians within each appointment track.

Q3

Pay Dispersion

P90 ÷ P10 ratio per agency to quantify internal inequality, benchmarked against the median ratio.

DC Pay is Tightly Centered at $100K



INTENT

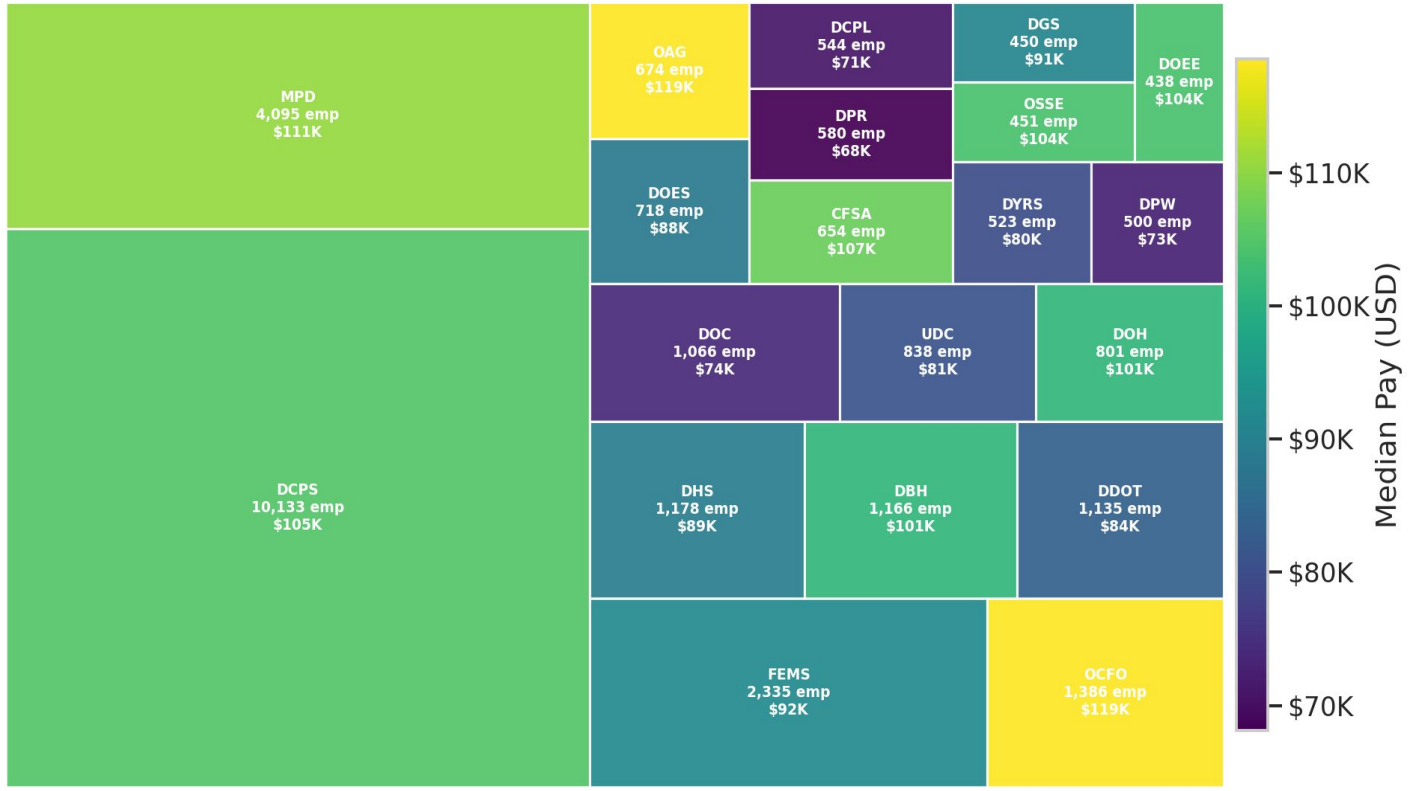
Establish the global shape of compensation before slicing it by group.

INSIGHT

Mean ($101.5K) and median ($100.3K) are nearly identical — unusual for a salary distribution. The right tail extends to $400K but contains too few employees to skew the mean. This is the empirical fingerprint of a capped, grade-driven pay system.

Three Agencies Carry Half the Workforce

DC Workforce Treemap — Size by Headcount, Color by Median Pay (Top 20 Agencies)



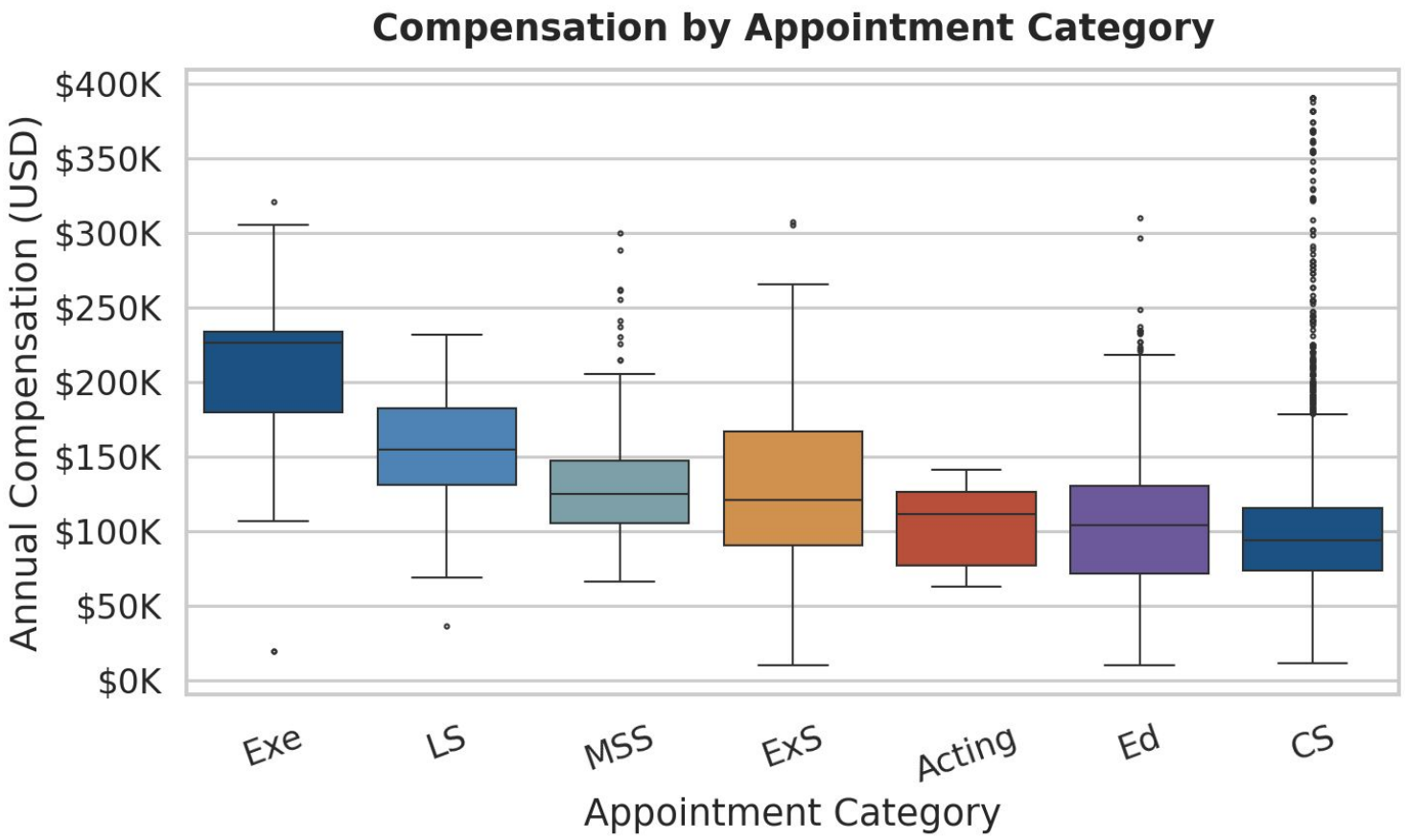
INTENT

Encode agency size by area and median pay by color. One picture exposes both structural concentration and pay variation simultaneously.

INSIGHT

DCPS alone is 28% of the cleaned dataset. Together, DCPS + MPD + FEMS account for nearly half of all DC public employees. Color reveals OCFO and OAG as small-but-high-paid outliers in a sea of larger, mid-pay agencies.

Appointment Category is the Strongest Pay Predictor



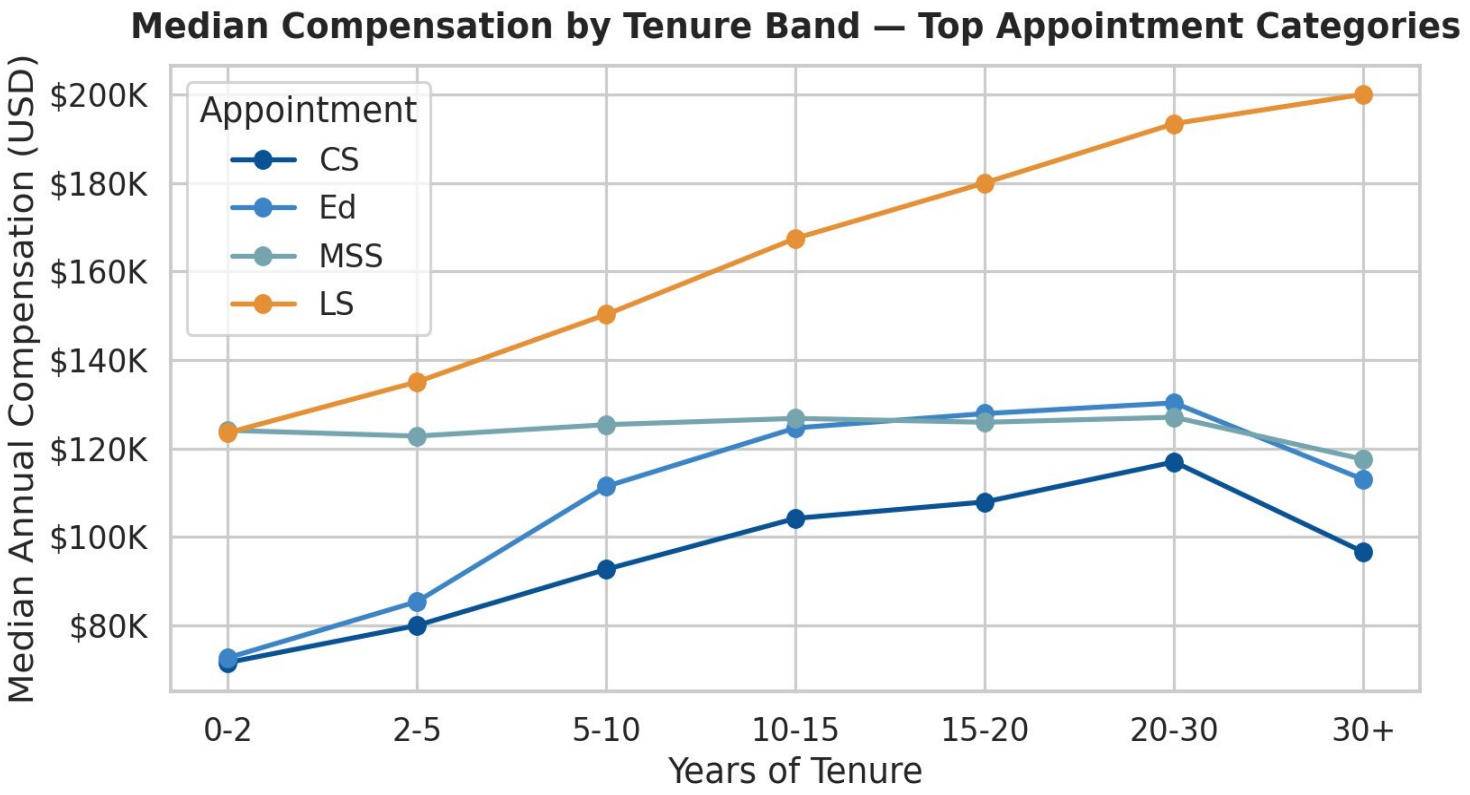
INTENT

Compare the seven appointment categories side-by-side using box plots, ordered by median.

INSIGHT

Clean ordering: Exe > LS > MSS > ExS > Acting ≈ Ed > CS. The Career Service category dominates by headcount but anchors the bottom — though its outliers reach $400K, suggesting senior directors who pre-date the modern Executive appointment type.

Each Career Track Has a Different Pay Arc



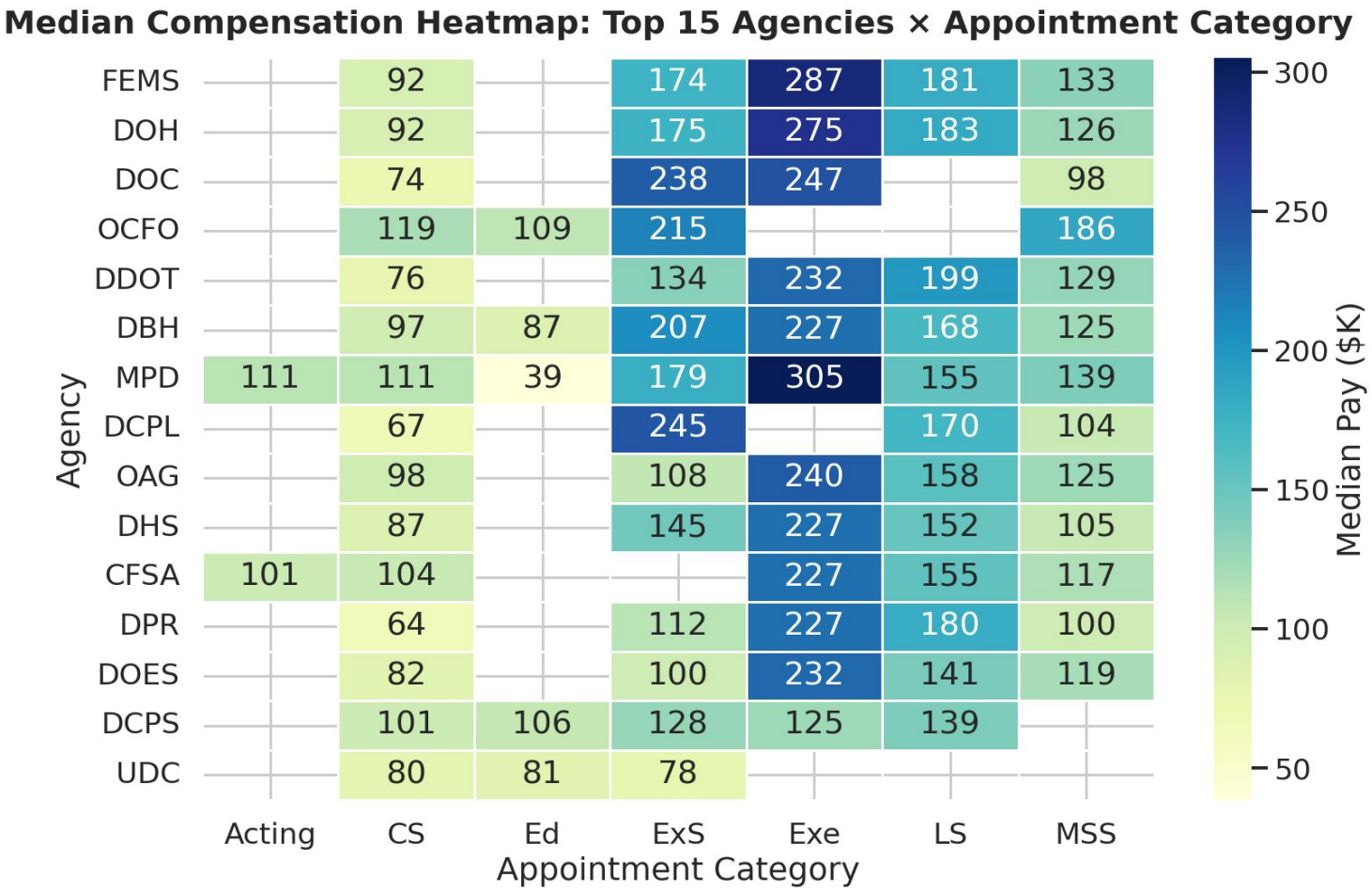
INTENT

Disaggregate the global tenure-pay correlation (r = 0.293) into within-category curves to expose differing career-arc shapes.

INSIGHT

Legal Service (LS) doubles pay across a career — $125K to $200K+. Management Supervisory (MSS) is flat from year one — pay is set on entry. Career Service climbs steadily, peaking at 20-30 years. Educational Service plateaus mid-career.

Cross-Section: Where the High Pay Concentrates



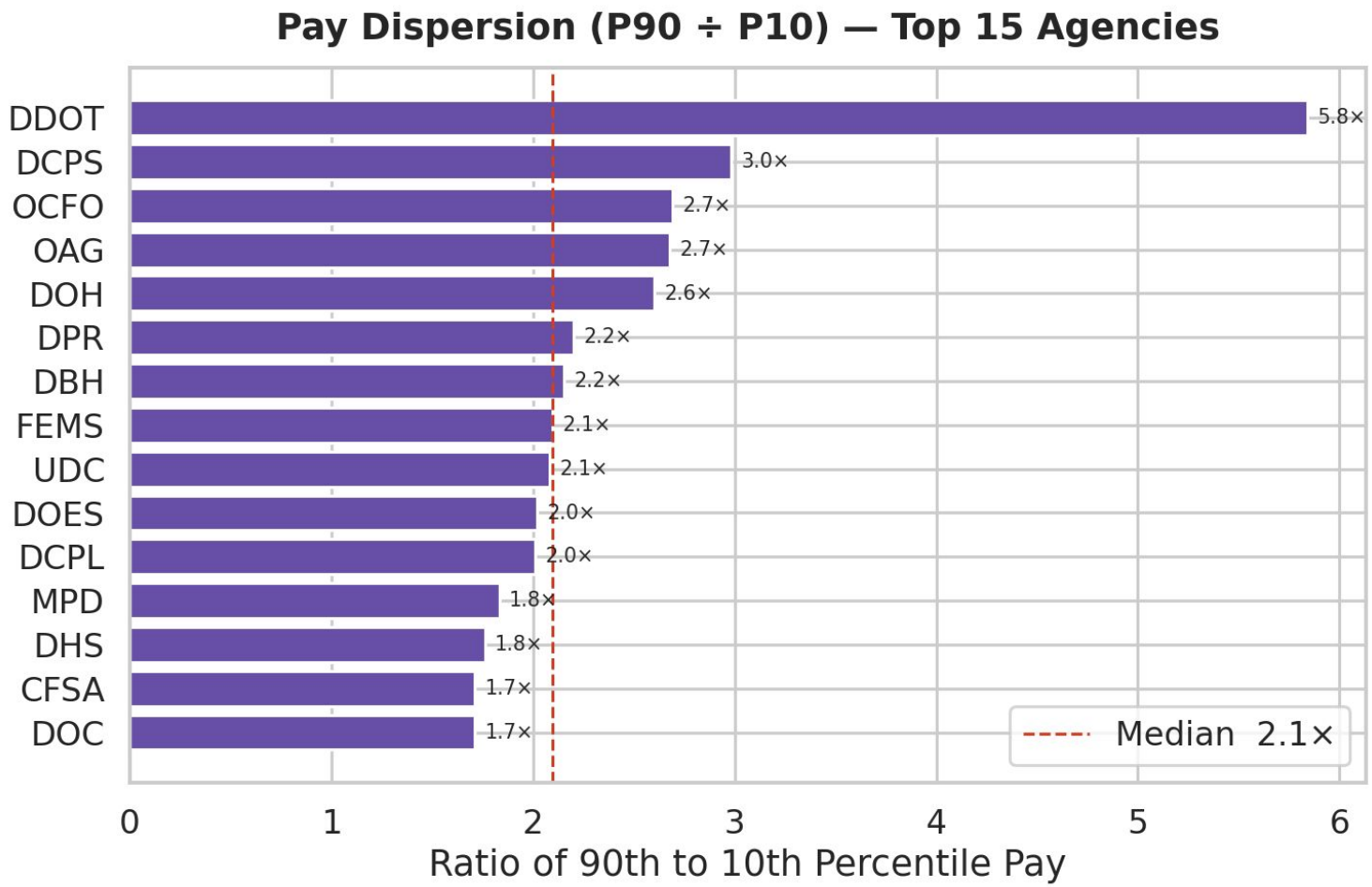
INTENT

Cross-tabulate the two strongest grouping variables (agency × appointment) to expose interaction effects invisible in either margin alone.

INSIGHT

OCFO Career Service medians ($119K) beat CS medians at almost every other agency. MPD's Executive tier tops $300K, dwarfing DCPS's $125K — different missions, different executive structures. Empty cells are themselves information about how each agency is organized.

DDOT is the Most Internally Unequal Agency



INTENT

Quantify within-agency pay inequality using the P90 ÷ P10 ratio. A ratio of 2.0× means top-decile pay is at least double bottom-decile pay.

INSIGHT

DDOT (5.8×) is a striking outlier — it mixes professional engineers and senior planners with a large field crew of laborers. DCPS (3.0 ×) and OCFO (2.7×) follow. DOC and CFSA sit at 1.7× — dominated by a single skilled-but-not-executive role profile.

What the data revealed

28%

of the cleaned workforce works for DCPS alone — nearly 1 in 3 DC employees teaches in public schools.

$100K

is the global median; mean and median are nearly identical, signaling a tightly capped grade system.

0.293

Pearson correlation between tenure and pay — modest pooled, but very different by appointment track.

5.8×

is DDOT's P90/P10 pay ratio — almost 3× the median agency, driven by a two-tier engineer/laborer mix.

CONCLUSION

DC's public workforce is more compressed and more education-driven than a casual look would suggest.

Pay rises with tenure, but the slope depends entirely on the track you're on. Pay dispersion within agencies ranges from 1.7× (DOC, CFSA) to 5.8× (DDOT) — the agencies with the widest spreads are those mixing credentialed professionals with operational workforces. Appointment category — not agency, not tenure — is the dominant axis of pay variation.

Thank you.

Questions welcome.

LIMITATIONS

- Cross-sectional snapshot — no individual trajectories.
- COMPRATE is annual rate; part-time normalization may be imperfect.
- No demographic columns — pay equity by race/gender unanswerable here.

REFERENCES

- DC Open Data — DC Public Employee Salary Dataset
- DC DCHR — Classification & Compensation framework
- pandas · seaborn · matplotlib · plotly express